Jaguar Re-Identification Report

TeamIOTA - Pranav Birari, Vedant Patil

EDA EXPERIMENTS

EDA experiments explored background reliance, embedding variance, class imbalance, and hard pair confusion across 31 jaguar identities. These experiments focused on understanding why the model succeeds or fails at the identity level rather than optimizing leaderboard performance.

#	Experiment	W&B Run
8	Background Reliance Analysis	background-reliance
9	Intra-Identity Embedding Variance	intra-identity-variance
15	Class Imbalance vs Per-Identity AP	class-imbalance-ap-correlation
16	Hard Pair Confusion Analysis	hard-pair-confusion-analysis

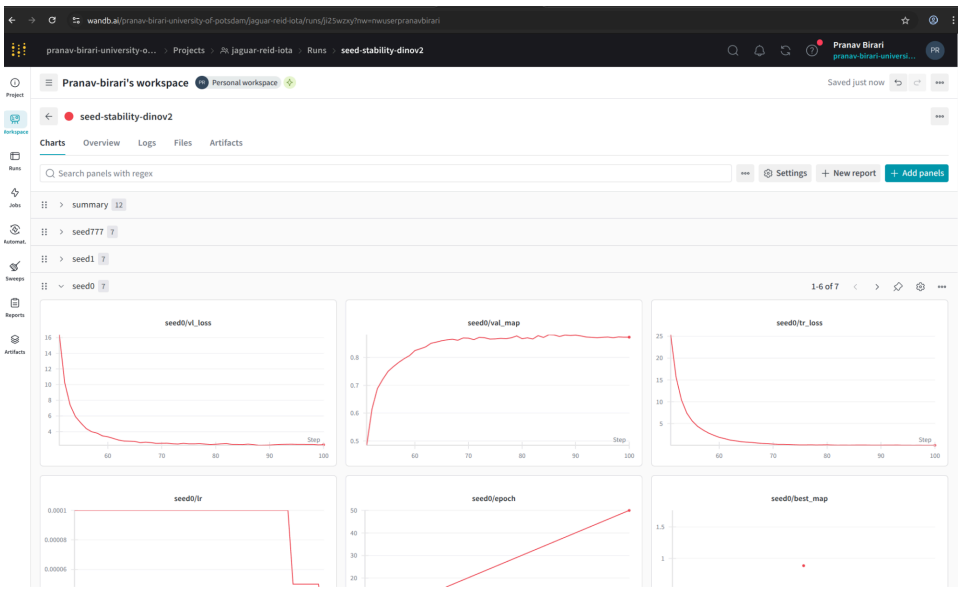
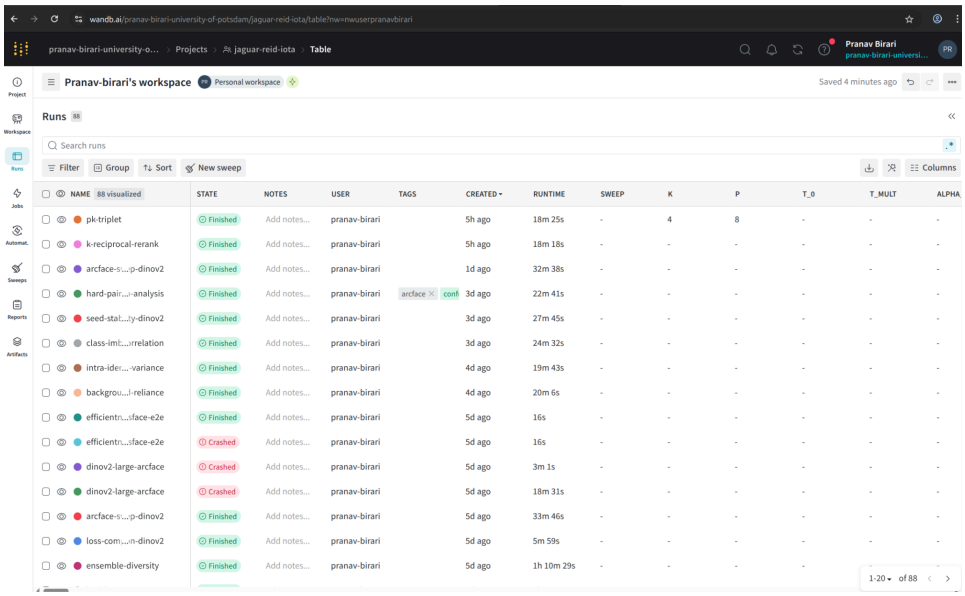
LEADERBOARD EXPERIMENTS

Leaderboard experiments covered backbone selection, loss function comparisons, ensemble fusion, inference-time techniques (TTA, re-ranking), PK sampling, and ArcFace hyperparameter tuning across 12 experiments. DINOv2-ViT-L/14 emerged as the strongest backbone despite not being wildlife-specific

#	Experiment	W&B Run	mAP	Score
1	Head-Only vs End-to-End Training	exp8-megadesc-arcface	0.7927	0.755
2	MobileNetV3 vs MegaDescriptor	mobilenetv3-arcface-backbone-comparison	0.7795	0.719
3	Loss Function Comparison (MegaDescriptor)	loss-comparison-q3	0.8020	0.772
4	Backbone Comparison (5 backbones)	backbone-comparison	0.8447	0.835
5	Ensemble Fusion (4 backbones)	ensemble-diversity	0.8358	0.846
6	TTA & Average Query Expansion	tta-aqe-q6	0.7760	0.760
7	PK Sampling & Batch-Hard Triplet Loss	pk-triplet-q7	0.7793	0.747
10	Swin Transformer Backbone	swin-arcface-local	0.8145	0.789
11	K-Reciprocal Re-Ranking	k-reciprocal-rerank-q8	0.7892	0.762
12	8-Loss Comparison on DINOv2	loss-comparison-dinov2	0.8270	0.803
13	ArcFace Hyperparameter Sweep (DINOv2)	arcface-sweep-dinov2	0.8631	0.864★
14	Seed Stability Analysis (7 seeds)	seed-stability-dinov2	0.8818	0.868★

Kaggle #1 -> W&B Run - seed-stability-dinov2
Kaggle #2 -> W&B Run - arcface-sweep-dinov2

Team Contribution Statement

- Pranav Birari led Experiments 3, 4, 5, 8, 11, 12, 13, and 14 covering loss function comparison, backbone comparison, ensemble fusion, background reliance analysis, k-reciprocal re-ranking, eight-loss comparison on DINOv2, ArcFace hyperparameter sweep, and seed stability analysis.
- Vedant Patil led Experiments 1, 2, 6, 7, 9, 10, 15, and 16 covering head-only vs end-to-end training, MobileNetV3 backbone comparison, TTA and query expansion, PK sampling and triplet loss, embedding variance analysis, Swin Transformer backbone, class imbalance analysis, and hard pair confusion analysis.
- Report writing, result interpretation, and Kaggle submissions were done together.

W&B Workspace